

19. If $z = -\sqrt{3} - i$, then $z^3 =$ _____.

(A) $8i$.

(B) $-8i$.

(C) $-7i$.

(D) $7i$.

20. If $z = 2 + \pi i$, then $e^z =$ _____.

(A) $-3e^2$.

(B) $2e^2$.

(C) $2e^3$.

(D) $-e^2$.

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FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS-UG)

Mathematics

MTS 5B 06—BASIC ANALYSIS

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer at least ten questions.
Each question carries 3 marks.
All questions can be attended.
Overall Ceiling 30.*

1. Is the union of two disjoint denumerable sets denumerable ?
2. If $a, b \in \mathbb{R}$ with $ab = 0$, then prove that either $a = 0$ or $b = 0$.
3. If $a \in \mathbb{R}$ is such that $0 \leq a < \varepsilon$ for every $\varepsilon > 0$, then show that $a = 0$.
4. Find all real numbers x satisfying the inequality $x^2 > 3x + 4$.
5. If $0 < c < 1$, then show that $0 < c^2 < c < 1$.
6. If x and $y \in \mathbb{R}$ with $x < y$ prove that there exists an irrational number z such that $x < z < y$.
7. State characterization theorem for intervals.
8. Test the convergence of $\left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \dots\right)$.
9. Show that every convergent sequence is a Cauchy sequence.
10. Define Supremum of a set and give example of a set which has no Supremum.
11. What can be said about the complex number z if $z = -\bar{z}$.
12. Find modulus of the complex number $z = -9i$.
13. Find real and imaginary parts of the complex function $f(z) = \bar{z}$ as functions of r and θ .
14. State nested interval property.
15. Write the equation of (a) a closed disk of radius ρ centred at z_0 ; (b) equation of a circle with centre z_0 and radius ρ .

(10 × 3 = 30 marks)

Turn over

Section B

*Answer at least five questions.
Each question carries 6 marks.
All questions can be attended.
Overall Ceiling 30.*

16. State and prove Cantor's theorem.
17. Prove that there does not exist a rational number r such that $r^2 = 2$.
18. Solve the inequality $|2x - 1| \leq x + 1$.
19. Let S be a non-empty set in $\mathbb{R}$, that is bounded above. Prove that $\text{Sup}(a + S) = a + \text{Sup } S$.
20. State and prove Archimedean property.
21. Prove that a sequence in $\mathbb{R}$ can have at most one limit.
22. Find the image of the half plane $\text{Re } z \geq 2$ under the mapping $W = iZ$.
23. Prove that $|z_1 - z_2| \geq ||z_1| - |z_2||$.

(5 × 6 = 30 marks)

Section C

*Answer any two questions.
Each question carries 10 marks.*

24. (a) State and prove Arithmetic-geometric inequality.
- (b) Let $a, b, c \in \mathbb{R}$. Then if $ab < 0$ then show that either $a > 0$ and $b < 0$ or $a < 0$ and $b > 0$.
- (c) If $1 < C$, then show that $1 < C < C^2$.
25. (a) Prove that every contractive sequence is a Cauchy sequence.
- (b) Prove that if a sequence X of real numbers converges to a real number x , then any subsequence of X also converge to x .
26. (a) The polynomial equation $x^3 - 7x + 2 = 0$ has a solution between 0 and 1. Use an approximate contractive sequence to calculate the solution correct to 4 decimal places.
- (b) Show that $\lim_{n \rightarrow \infty} \left(\frac{1}{n^n} \right) = 1$.
27. (a) Find an upperbound for $\left| \frac{1}{z^4 - 5z + 1} \right|$ if $|z| = 2$.
- (b) Find the image of the vertical strip $2 \leq \text{Re } Z < 3$ under the mapping $f(Z) = 3Z$.
- (c) Find the domain of $f(z) = \frac{iz}{|z| - 1}$.

(2 × 10 = 20 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021**(CBCSS—UG)****Mathematics****MTS 5B 05—THEORY OF EQUATIONS AND ABSTRACT ALGEBRA****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5B 05—THEORY OF EQUATIONS AND ABSTRACT ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. If H_1 and H_2 are two right cosets of subgroup H then :

(A) $H_1 \cap H_2 = \phi$ $H_1 = H_2$.

(B) $H_1 \cap H_2 \neq \phi$.

(C) $H_1 \cup H_2 = \phi$.

(D) $H_1 \neq H_2$ and $H_1 \cap H_2 \neq \phi$.

2. (i) Every isomorphic image of a cyclic group is cyclic.

(ii) Every homomorphic image of cyclic group is cyclic.

(A) Both (i) and (ii) are true.

(B) Both (i) and (ii) are false.

(C) (i) is true only.

(D) (ii) is true only.

3. E is set of integers under ordinary addition and multiplication, then E is a ring, E is also a _____.

(A) Commutative ring.

(B) Integral domain.

(C) Both (A) and (B).

(D) None of these.

4. The set of all rational numbers of the form $3^m \cdot 6^n$; m, n are integers :

(A) Forms a group under multiplication.

(B) Forms a group under addition.

(C) Forms a group under division.

(D) None of the above.

5. A group contains two elements a and b such that $|a| = 4$, $|b| = 2$, and $a^3 \cdot b = ba$. Then $|ab| =$

(A) 2.

(B) 4.

(C) 6.

(D) 8.

6. Let G_n be a cyclic group of order n . Which of the following direct product is not cyclic ?

(A) $G_{22} \times G_{31}$.

(B) $G_{222} \times G_{333}$.

(C) $G_{17} \times G_{11}$.

(D) $G_{17} \times G_{11} \times G_5$.

7. What is the largest order of an element in the group of permutations of 5 objects ?
- (A) 5. (B) 6.
(C) 12. (D) 60.
8. The set of all generators of a cyclic group $G = \langle a \rangle$ of order 8 is :
- (A) $\{a^2, a^4, a^6\}$. (B) $\{a^1, a^3, a^5, a^7\}$.
(C) $\{a^4, a^8\}$. (D) $\{a^3, a^5, a^7\}$.
9. Number of self-inverse elements of D_4 :
- (A) 0. (B) 2.
(C) 4. (D) 6.
10. Identity element of the group R under addition is :
- (A) 1. (B) 2.
(C) 0. (D) -1.
11. $\{1, i, -i, -1\}$ is _____.
- (A) Semigroup. (B) Subgroup.
(C) Cyclic group. (D) Abelian group.
12. Let A, B be non-empty sets and $f : A \rightarrow B$ be a permutation. Then _____.
- (A) f is bijective and $A = B$. (B) f is one one and $A \neq B$.
(C) f is bijective and $A \neq B$. (D) f is onto and $A \neq B$.
13. The number of elements in $A_6 =$ _____.
- (A) 6. (B) 720.
(C) 360. (D) 2^6 .
14. Let H be a subgroup of a group G and $a, b \in G$. Then $b \in aH$ if and only if :
- (A) $ab \in H$. (B) $ab^{-1} \in H$.
(C) $a^{-1}b \in H$. (D) None of these.

15. If G be a cyclic group of order 8 with generator x then another generator of G be :

- (A) x^5 . (B) x^4 .
(C) x^6 . (D) x^2 .

16. Let A_6 be the group of even permutations of 6 distinct symbols. Then the number of elements of order 6 in A_6 is :

- (A) 0. (B) 1.
(C) 3. (D) 4.

17. If G is commutative group then $(ab)^n = \dots \forall n \in \mathbb{Z}$.

- (A) ab . (B) ba .
(C) $a^n b^n$. (D) $a^n b^m$.

18. If $(x - a)$ is a factor of $f(x)$ then $f(a)$:

- (A) a . (B) 0.
(C) 1. (D) $a - 1$.

19. Sum of the roots of the equation $x^5 - 5x^3 + x + 1 = 0$ is given by :

- (A) 0. (B) 5.
(C) -1. (D) None of these.

20. A polynomial equation in x of degree n always has :

- (A) n distinct roots. (B) n real roots.
(C) n imaginary roots. (D) At most one root.